Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/11/2005
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Project Title:	Coupled Radiation/Thermophoresis Effects in Sooting Microgravity Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	36849	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Nayagam, Vedha (National Center for Microgravity Research) Sunderland, Peter B. (National Center for Microgravity Research)		
Grant/Contract No.:	NCC3-864		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	Carbonaceous soot particles formed in hydrocarbon flames are highly efficient radiators of heat. Because of this, the particles act as heat sinks in the high temperature combustion gases, and the gas will respond by conducting heat into the particles. This conductive heat flux, in turn, results in a transport of particles by thermophoresis, which can drive the particles towards each other and into contact. Essentially, this coupled action of radiative emission and thermophoretic transport creates a coagulation mechanism in the flame. Recent experimental observations in sooting microgravity diffusion flames, as well as simple analytical models, indicate that this mechanism could result in a destabilization of the soot field and the formation of very large (super-micron) soot aggregates. This research project will examine, through a combined experimental and theoretical/numerical program, the effects of combined radiation/thermophoresis interactions on the evolution of aerosols in high temperature gases. The objective of our proposed work is determine the conditions under which coupled radiation and thermophoresis leads to large aggregate formation in flames. The research will be undertaken in three concurrent tasks. The first task will be an experimental examination of radiation/ thermophoresis effects in spherical, hydrocarbon/air diffusion flames that are formed in microgravity environments (using the 2.2 s drop tower). Microgravity is needed because 1) it provides a flame residence time that is comparable with our estimates of the characteristic coagulation times resulting from radiation/thermophoresis coupling; 2) it eliminates particle aggregation by fluid shear; and 3) large-scale aggregates are known to form in such flames. We will test the radiation/thermophoretic mechanism by controlling the radiation environment of the flame. Specifically, we will project a source of radiation into the soot-aggregating regions of the flame, the intensity of which will be sufficiently large so that radiation emission from the soot becomes balanced by radiative absorption. If radiation/thermophoresis is driving large-aggregate formation in the flame, then a balancing of radiative emission with absorption will arrest the aggregation. We will use direct imaging methods to determine the extent to which the radiative field alters the coagulation dynamics of the particles. The second task of the proposed research is to develop a numerical model for predicting the effects of radiation/thermophoretic coupling on soot volume fraction in diffusion flames. We will base the model on the spherical, microgravity diffusion flame used in the experiment, and employ simplified, one-step kinetic models to describe soot formation and oxidation. However, the model will be fully 3D to account for the 3D restructuring of the soot field by the coupled mechanisms. The goals of this effort will be to identify conditions under which the soot field becomes destabilized (i.e., increasingly nonuniform) by radiative emission and thermophoretic transport. The third task of the proposed research will be to develop accurate analytical models of the radiative emission, hydrodynamic drag, and thermophoretic properties of realistic, fractal-like aggregates of spheres. Such information is needed to accurately assess the effects of radiation/thermophoretic aggregation of soot (in particular) and overall convective/diffusive soot transport in flames (in general), yet a detailed understanding of these properties for soot particles is lacking. Observations of diffusion flames in microgravity flames, as well as simple theoretical models, indicate that the radiation/ thermophoresis mechanism can lead to a snowballing of aggregate growth (i.e., 'superaggregate' formation) in situations of sufficiently long residence time. A detailed experimental and theoretical examination is needed to fill in the understanding of this mechanism, i.e., under what flame conditions and time scales will it lead to soot field instability. Furthermore, the interaction of particle radiation and thermophoresis has relevance beyond that of 'natural' soot formation in flames. Unlike other coagulation mechanisms, radiation/thermophoresis could potentially be controlled by manipulation of the radiative field -- much as we are proposing to do in our experiments. This mechanism could therefore be exploited to either accelerate or arrest the coagulation of aerosols, and such an ability would have immense practical importance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the tasks in the project is to provide improved models of radiation absorption by soot aerosols. These methods will be of benefit to the prediction of the effects of soot on atmospheric radiation forcing, as well as the modelling of heat transfer from flames.
Task Progress:	Ground--based experiments were conducted in the summer of 2004, with the objective of identifying soot coagulation effects resulting from radiation and thermophoretic interactions. The experiments utilized a coflow, sooting acetylene/air diffusion flame, with the soot plume maintained at a high temperature by a coannular, premixed methane/air flame. Gelation of the soot emitted by the acetylene flame was observed, although the rate of gelation was not apparently affected by the temperature environment of the soot. Furthermore, individual, large--sized soot particles did not appear to be attracted to each other, as predicted by the theory, when the slow--moving flowfield of the plume brought the particles into close proximity. The experiments were therefore inconclusive with regard to identifying the presence of the hypothesized radiation--thermophoretic coagulation mechanism in sooting diffusion flames. Analytical work has been completed to predict the absorption cross sections of fractal--like soot aggregates. The model uses an electrostatics formulation (Rayleigh limit) for neighboring spheres. Results show that the absorption cross section of aggregates can be 20--50 percent larger than that of the isolated primary spheres in the near to mid IR wavelengths. This effect is uniquely due to the aggregated sphere structure of the particles. A power--law formula was developed to provide a simple method of predicting the absorption cross sections of aggregates as a function of refractive index and number of primary spheres in the aggregate. Analytical/numerical work has also been completed on a detailed model of momentum and heat transfer from fractal--like aggregates in the free molecular Knudsen regime. The objective of this work, which forms the basis of the MS thesis of a graduate student supported by the project, is to provide a detailed prediction of the thermophoretic forces between radiatively--heated or cooled aggregates and estimate the effect of these forces on the coagulation rates of the aggregates. This work complements the near--continuum model previously developed by the author.
Bibliography Type:	Description: (Last Updated: 04/11/2005)
Articles in Peer-reviewed Journals	Mackowski, D. W. "Monte--Carlo simulation of hydrodynamic drag and thermophoresis of fractal aggregates of spheres in the free--molecule flow regime" volume 36 (available online) , Jan-2005

Presentation	Mackowski, D. W. "A simplified model to predict the effects of aggregation on the absorption properties of soot particles" Eight International Symposium on Light Scattering by Nonspherical Particles, Salobrena, Spain May-2005